

Name: Namam Babbar

Roll No: 2223604

Subject: UC D

Enrollment No: MITU22BTCS0467

Assignment - 3

- Q1. Discuss Device specific interaction with respect to computing devices.
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- i) Tailored interaction for each device type (smart phones, tablets, desktops, etc) based on its design & functionality.
 - ii) Different input methods like touch (smart-phones), keyboard/mouse (desktops) and voice commands.
 - iii) Device interaction adapts to its usage environment like quick interactions for mobile and detailed work for desktops.
 - iv) Applications are optimized for device specific constraints like battery life, processing power, and screen size.
 - v) Multi-device synchronization allows seamless transitions b/w devices (e.g. from phone to desktops).

Q2. List and explain seven stages of interaction model short and proper points.

→ 7 stages of Interaction Model :-

i) Forming the Goal :

- The user identifies what they want to achieve through interaction with the system or device.

ii) Forming the intention :

- The user decides on the specific action they need to take to accomplish the goal.

iii) Specifying the Action :

- The user translates the intention into a sequence of actions, deciding what commands or inputs are needed.

iv) Executing the Action :

- The user performs the physical action, such as clicking a button or typing.

v) Perceiving the System State :-

- The user observes the system's response or feedback to the action taken.

vi) Interpreting the System State :-

- The user interprets the system feedback to understand whether their action was successful.

vii) Evaluating the Outcome :-

- The user compares the result to their original goal to determine if the goal was to meet or if further action's needed.

Q3. Write short note on: Voice Interaction

i) Hands-Free Operation

- Allows users to control devices and perform tasks using voice commands without physical input (e.g. Siri, Alexa).

ii) Natural Language Processing (NLP)

- Uses NLP technology to understand and interpret spoken language, making interaction more conversational.

iii) User Convenience

- Offers ease of use for people with disabilities providing an alternative to traditional input methods like typing.

iv) Accessibility

- Enhances accessibility for people with disabilities providing an alternative to traditional input methods like typing.

v) Challenges

- Can struggle with accents, background noise, and recognizing context, leading to occasional errors in interaction.

Q4. List various interaction styles. Explain any one of them.

→ Various Interaction Styles

i) Command Line Interface

ii) Menu Driven Interface

iii) Graphical User Interface

iv) Form-Fill interaction.

v) Direct Manipulation

vi) Natural Language Interface

vii) Voice Interaction

* Explanation of Direct Manipulation

i) Direct Manipulation involves interacting with objects on the screen directly, such as dragging and dropping files, resizing windows, or clicking on icons.

ii) It provides immediate feedback, making the interface intuitive and easy to understand.

iii) This style is common in modern operating systems and helps users perform tasks efficiently with minimal learning curve.

Q5. What are Audio-Visual Interface?
Explain briefly.

→ i) Definition

• An audio-visual interface combines both sound (audio) and visual elements (images, videos, animations) to enable interactions.

ii) Components

• Audio - Spoken instructions, sound effects, or music for communication or feedback.

• Visual - Graphical elements like buttons, icons, videos, or images displayed on the screen.

iii) Usage

Commonly used in multimedia applications, video games, virtual reality, and training simulators.

to enhance user experience by engaging multiple senses.

iv) Advantages

- Make interactions more intuitive and engaging
- Helps convey complex information in an easy-to-understand manner.

v) Challenges

Requires good design to balance both audio and visual elements for a smooth and effective user experience.

~~Q6.~~ List and explain seven stages of interaction model in short and proper points.

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Q6. List and explain steps for creating user scenarios.

→ i) Identifying Target Users:

- Define user demographics, roles and needs.

ii) Understand User Goals:

- Determine what the users aim to achieve with the system.

iii) Create Context

- Set the environment, time, and situation of the interaction.

iv) Outline User Actions:

- List the steps the user takes to reach their goal.

- v) Consider System Response :
- Describe potential user mistakes and recovery options.
- vi) Identify Possible Errors :
- Predict potential user mistakes and recovery options.
- vii) Refine and validate :
- Review, test and adjust the scenario with real users for accuracy.

Simple